Figure 6: The image we want to detect

We can see that the cropped image 'black ops' is matched with the portion of the full image, and the area is outlined by a yellow rectangle.

Gradients and edge detection

For detecting gradients and edge, type:

```
import cv2
import numpy as np
cap=cv2.VideoCapture(0)
while True:
     _, frame=cap.read()
     a= cv2.Laplacian(frame,cv2.CV_64F)
     x= cv2.Sobel(frame,cv2.CV_64F,1 ,0,
ksize=5)
     y= cv2.Sobel(frame,cv2.CV_64F,0 ,1,
ksize=5)
     edge= cv2.Canny(frame, 100, 200)
     cv2.imshow('original',frame)
     cv2.imshow('laplacian',a)
     cv2.imshow('sobelx',x)
     cv2.imshow('sobely',y)
     cv2.imshow('edge',edge)

     k=cv2.waitKey(5) & 0xFF
     if k == 27:
     break
cv2.destroyAllWindows()
cap.release()
```

cv2.Laplacian converts the image into the gradient, while *cv2.Sobel* converts it into a horizontal and vertical gradient. We use *cv2.CV_64F* as a standard label.

The output is shown in Figure 7.

Edge detection is pervasive in many applications such as fingerprint matching, medical diagnoses and licence plate detection. These applications basically highlight the areas where image intensity changes drastically, and ignore everything else.

Edge detection is also used in self-driving cars for lane detection.

Other features of OpenCV include motion detection, intrusion detection, homography, corner detection, colour filtering, thresholding, image arithmetic, etc.

Figure 7: Laplacian

Figure 8: Vertical gradient

Figure 9: Horizontal Gradient

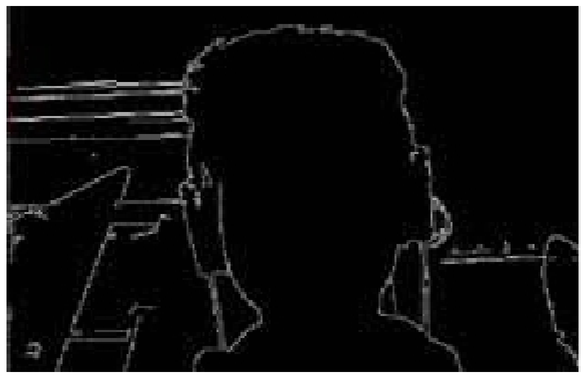

Figure 10: Edge detection

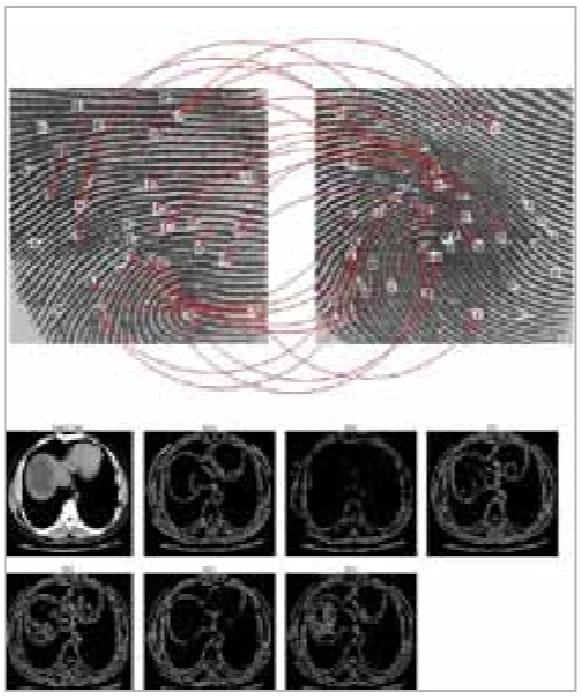

Figure 11: Fingerprint matching and edge detection of an embryo in the mother's womb (Source: *https://www.quora.com/What-are-the-applications-of-edge-detection/answer/Ravindra-Sadaphule*

Figure 12: Lane detection of roads using OpenCV (Source: *https://www.youtube.com/watch?v=xdiekchp-Uc*)

Statistical machine learning libraries used by OpenCV include Naive Bayes classifier, K-nearest neighbour algorithm, decision-tree learning, meta algorithm, random forest, support vector machine, and deep and convolutional neural networks.

Some popular applications of OpenCV are:

- Driver drowsiness detection (using a camera in a car) by alerting the car driver with a buzz or alarm.
- Counting vehicles on highways (can be segregated into buses, cars, trucks) along with their speeds.
- Anomaly detection in the manufacturing process (the odd defective product).
- Automatic number plate recognition (ANPR) to trace vehicles and count the number of passengers.

OpenCV is also used for data imaging to provide better diagnosis and treatment for a range of diseases. END

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The author is interested in the latest open source technologies and is an active contributor to the community.